AI-based re-identification exposes privacy risk of behavioral data. A case for synthetic data

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The Re-Identification of Netflix Data

Netflix releases "anonymized" data on 2006-10-02

- 470K users, 18K movies, 100M ratings
- only subset of customer base
- no customer information
- some random noise to dates and ratings

released Netflix data

users_id	movie	date	rating (1-5)
7	Blast	2003-11-21	4
7	The Wedding Planer	2003-11-21	5
7	Collateral Damage	2004-05-05	3
7	S.W.A.T.	2004-05-05	5
7	Whale Rider	2004-05-20	1
...	...	...	...
2649429	Scorched	2005-09-27	3
2649429	The Last Samurai	2005-10-10	4
2649429	The Longest Yard	2005-10-18	2
2649429	Bad Boys	2005-12-23	5
2649429	The Village	2005-12-23	5

Paper on Re-Identification on 2006-10-18

- fuzzy linkage attack
- leveraged public IMDB data as auxiliary

How To Break Anonymity of the Netflix Prize Dataset

Arvind Narayanan and Vitaly Shmatikov
The University of Texas at Austin

auxiliary IMDb data

user	movie	date	rating (1-10)
melissa.ricks	The Wedding Planer	2003-11-22	9
TylerDurden	The Wedding Planer	2003-12-02	8
...	...	...	...
melissa.ricks	Whale Rider	2004-05-21	2
...	...	...	...

Aftermath

1. class action lawsuit against Netflix → undisclosed settlement
2. hardly any public sharing of behavioral data
3. privacy regulations adapted to linkage attacks

GDPR

https://gdpr-info.eu/issues/personal-data/

Personal data are **any information** which are **related to** an identified or **identifiable natural person**.

Recital 30 Natural persons may be associated with online identifiers provided by their devices [..] This may leave traces which [..] may be used to create profiles of the natural persons and identify them.

Recital 26 To determine whether a natural person is identifiable, account should be taken of all the means reasonably likely to be used, such as **singling out, either by the controller or by another person** to identify the natural person directly or indirectly. To ascertain whether means are reasonably likely to be used to identify the natural person, account should be taken of all objective factors, such as the costs of and the amount of time required for identification, **taking into consideration the available technology** at the time of the processing and technological developments.

AI-Based Re-Identification of Faces

?

Sylvester Stallone

...

Arnold Schwarzenegger

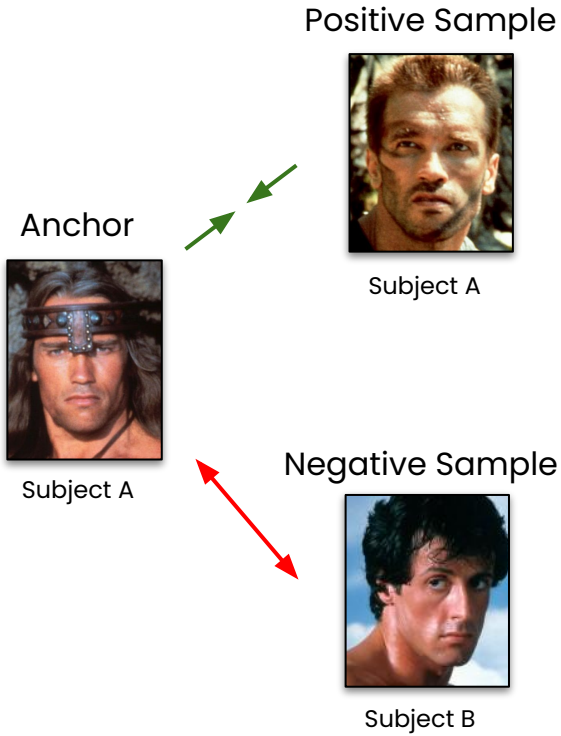
...

George Clooney

...

... ...

Learning Traits of Faces with Triplet Loss



Train Deep Neural Network to discriminate triplets of faces. This task will yield an embedding space representing the characteristic traits. (Schroff et al, 2015)

-0.23	0.39	0.92	-0.02	0.05	...	-0.24

→ Re-identification is then done via Nearest-Neighbor Search in that embedding space.

Learning Traits of Behavior with TL-RNN

WIRTSCHAFTS UNIVERSITÄT WIEN VIENNA UNIVERSITY OF ECONOMICS AND BUSINESS

Positive Sample

domain	pages_viewed	duration	day_of_week	hour_of_day
pinterest.com	1	1	Thu	15
etsy.com	2	1	Wed	1
americangreetings.com	3	2	Wed	16
lds.org	18	23	Fri	17
...	...	...	...	...

Subject #123

Anchor

domain	pages_viewed	duration	day_of_week	hour_of_day
lds.org	1	1	Tue	2
google.com	2	2	Tue	2
google.org	1	1	Tue	2
livestrong.com	1	1	Tue	2
amazon.com	28	22	Wed	15

Subject #123

Negative Sample

domain	pages_viewed	duration	day_of_week	hour_of_day
target.com	1	1	Tue	14
ksl.com	7	4	Wed	15
google.com	2	3	Fri	14
myway.com	2	4	Fri	22
...	...	...	...	...

Subject #789

Train Deep Neural Network to discriminate triplets of behavioral data. This task will yield an embedding space representing the characteristic traits of users. (Vamosi et al, forthcoming)

domain	pages_viewed	duration	day_of_week	hour_of_day
lds.org	1	1	Tue	2
google.com	2	2	Tue	2
google.org	1	1	Tue	2
livestrong.com	1	1	Tue	2
amazon.com	28	22	Wed	15

-0.23	0.39	0.92	-0.02	0.05	...	-0.24

→ Re-identification is then done via Nearest-Neighbor Search in that embedding space.

EFMD EQUIS ACCREDITED AACSB ACCREDITED ASSOCIATION OF MBAS ACCREDITED

Re-Identification Study

Attack Scenario

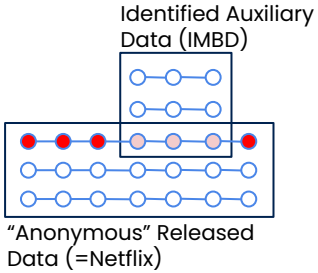
1. Organization releases a behavioral "anonymous" dataset D for period P1

2. Attacker obtains auxiliary behavioral data on user X for period P2

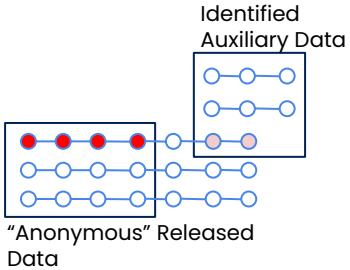
3. Attacker then attempts to re-identify user X within D via TL-RNN embeddings trained on D

   → Attacker reveals activities of user X within D

Linkage Attacks rely on an overlap of the data points of the released and the auxiliary data. A fuzzy match on these data points allows for re-identification.



Pattern Attacks do not require an overlap of the data points, but merely of the data subjects of the released and the auxiliary data. A fuzzy match on the behavioral patterns of these data points allows for re-identification.



Re-Identification Study

Dataset

- Comscore Web Browser Panel = continuous tracking of browsing behavior
- Release January to June data for 4,000 active “anonymous” panelists
- Attempt to re-identify panelists based on their observed July data → no overlap in data points

Subject #1 Jan-Jun
- expedia.com
- kayak.com
- google.com
- kayak.com
- ups.com
- usatoday.com
- ...

....

Subject #4000 Jan-Jun
- weather.com
- google.com
- usatoday.com
- google.com
- aol.com
- google.com
- ...

Subject #? Jul
- google.com
- google.com
- booking.com
- kayak.com
- cnn.com
- weather.com
- ...

Arnold Schwarzenegger

Re-Identification Study

Research Questions

1. Can we re-identify via pattern attack?
2. Can we protect with data perturbation?
3. Can we protect with data synthesis?

Re-Identification Study

0.025% (=1/4000) are re-identified

via random guess

Re-Identification Study

49.9% are re-identified

via TL-RNN based Pattern Attack

→ Re-Identification on behavioral traits possible

Re-Identification Study

65.6% are within Top 5 candidates

via TL-RNN based Pattern Attack

→ Re-Identification on behavioral traits possible

Re-Identification Study - Perturbation

26.6% are re-identified

despite 30% of data points being replaced*

→ Re-Identification is robust against Noise

*We replaced any data point with 30% probability with a data point from any other subject.

Highly destructive mechanism in an attempt to prevent re-identification.

Re-Identification Study - Perturbation

1.1% are re-identified

despite 60% of data points being replaced*

→ Re-Identification is robust against Noise

*We replaced any data point with 60% probability with a data point from any other subject.

Highly destructive mechanism in an attempt to prevent re-identification.

Re-Identification Study - Synthetization

Idea: Release **Synthetic Data** rather than (perturbated) Original Data

- Generative AI can provide representative datasets that strive to retain statistical properties
- no 1:1 link between actual and synthetic subjects → thus **no re-identification possible**
- BUT one still might leak information on individuals by memorization

How to Test Privacy of Synthetic Data?

- "Holdout-Based Fidelity and Privacy Assessment of Mixed-Type Synthetic Data" (Platzer et al., forthcoming)
- Require that synthetic subjects are NOT systematically closer to training subjects than to holdout subjects

Re-Identification Study - Synthetization

Empirical Privacy Test

- Split 4,000 subjects into 2,000 training and 2,000 holdout
- Generate 2,000 synthetic subjects based on 2,000 training
- Check whether synthetic are any closer to training than to holdout based on TL-RNN embeddings

Results

- Avg Distance to Training: 0.731
- Avg Distance to Holdout: 0.737
- 51.8% are closer to training - 48.2% are closer to holdout

dataset	DCR share
10% flip	98.5%
20% flip	89.9%
30% flip	70.6%
40% flip	62.6%
50% flip	58.7%
60% flip	58.1%
70% flip	57.6%
80% flip	55.4%
90% flip	57.5%
synthetic	51.8%

Summary

- Sharing of behavioral data is subject to GDPR
- AI-based re-identification on behavioral traits is possible
- Data perturbation does not protect your privacy
- Data synthesis can offer true anonymization